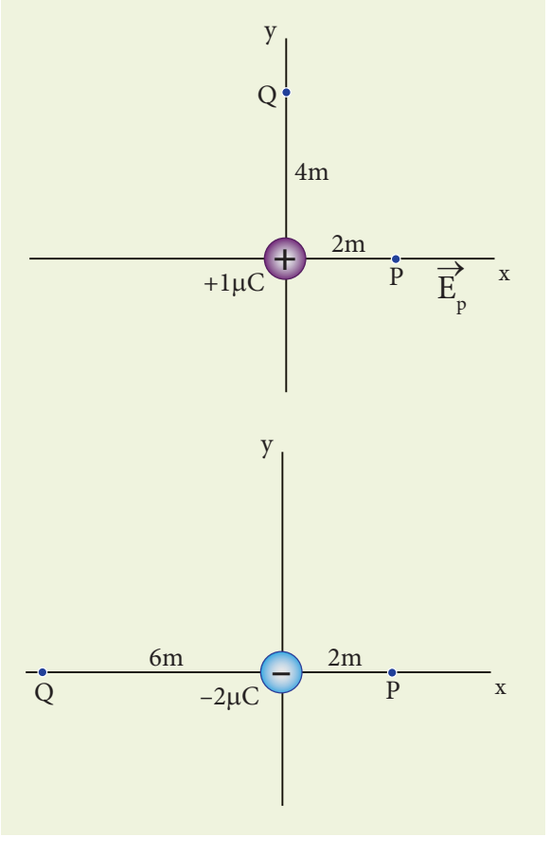
(a) A positive point charge +1 μC is placed at the origin

(b) A negative point charge −2 μC is placed at the origin



Solution

Case (a)

The magnitude of the electric field at point P is

$$E_P = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2} = \frac{9\times10^9\times1\times10^{-6}}{4}$$
$$= 2.25\times10^3\,\text{NC}^{-1}$$

Since the source charge is positive, the electric field points away from the charge. So the electric field at the point P is given by

$$\vec{E}_P = 2.25\times10^3\,\hat{i}\,\text{NC}^{-1}$$

For the point Q

$$\left|\vec{E}_Q\right| = \frac{9\times10^9\times1\times10^{-6}}{16} = 0.56\times10^3\,\text{NC}^{-1}$$

Hence $\vec{E}_Q = 0.56\times10^3\,\hat{j}\ \text{NC}^{-1}$

Case (b)

The magnitude of the electric field at point P

$$\left|\vec{E}_P\right| = \frac{kq}{r^2} = \frac{1}{4\pi\epsilon_0}\frac{q}{r^2} = \frac{9\times10^9\times2\times10^{-6}}{4}$$
$$= 4.5\times10^3\,\text{NC}^{-1}$$

Since the source charge is negative, the electric field points towards the charge. So the electric field at the point P is given by

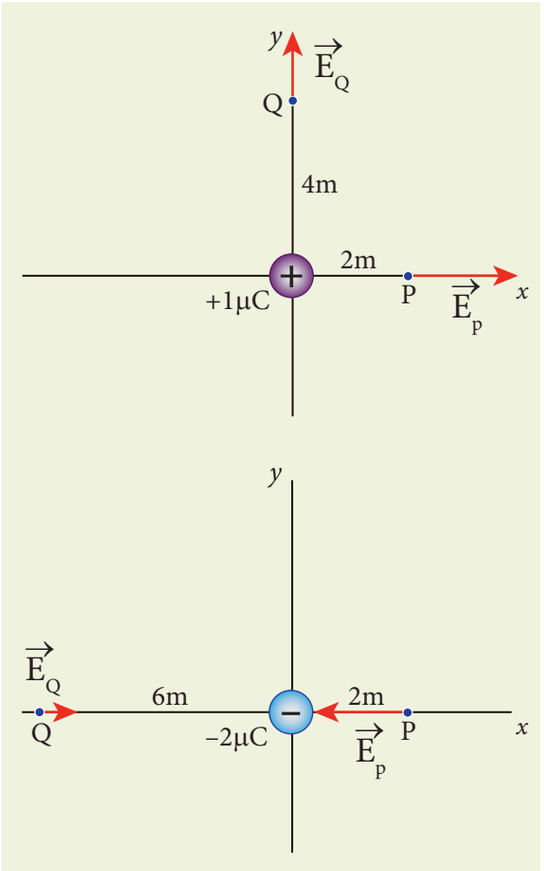
$$\vec{E}_P = -4.5\times10^3\,\hat{i}\ \text{NC}^{-1}$$

For the point Q, $\left|\vec{E}_Q\right| = \frac{9\times10^9\times2\times10^{-6}}{36}$

$$= 0.5\times10^3\,\text{NC}^{-1}$$

$$\vec{E}_Q = 0.5\times10^3\,\hat{i}\ \text{NC}^{-1}$$

At the point Q the electric field is directed along the positive *x*-axis.



12th Physics_Vol-1_Unit 1_English Medium.indd 14 08-12-2022 11:02:33